

Analysis of Complex Sample Survey Data

SURV/SURVMETH 701

Lecture Notes: Module 10

Logistic Regression Analysis for Complex Samples

Instructor: Yan Li

e-mail: yli6@umd.edu

Brief Review

- Linear regression for complex samples
- Difference from OLS
 - Goal: estimate relationships of variables in a **finite** population, using sample data
 - Variance estimation formulas appropriate for stratified cluster samples: Taylor series, Replication methods
- Similarity with OLS
 - Weighted least squares (WLS) estimation of parameters
 - Residual analysis for checking normality of residuals, homogeneity of variance, model fit
 - *t*-tests of individual parameters in regression
 - Wald tests or transformed F-tests of multi-parameter hypotheses

Logistic Regression Analysis of Complex Sample Survey Data

Outline of Topic

- The standard logistic regression model (in an SRS setting)
- The need for pseudo maximum likelihood estimation with complex sample designs
- Parameter estimation and variance estimation in the complex sample design setting

Software Applications / Examples

- Stata

Examples of Categorical Response Data

HRS 2006 N001: “Are you currently covered by Medicare health insurance?”

NHANES DIQ010: “Have you ever been told by a doctor or health professional that you have diabetes or sugar diabetes?”

NCSR SC26_4: “Did you ever use drugs or alcohol so often that it interfered with your responsibilities at work, at school or at home?”

Analytic Problem in Binary Logistic Regression Models

Response: $Y = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = f(x)$

Predictors: X

- 1) Categorical – Nominal (e.g., Region)
- 2) Categorical – Ordinal (e.g., Education)
- 3) Continuous (e.g., Weight in kg)
- 4) Mixture of Categorical and Continuous

Odds Ratio Statistic

$$\psi = \frac{Odds(A)}{Odds(B)} = \frac{P_{A1} / P_{A0}}{P_{B1} / P_{B0}} = \frac{P_{A1} \cdot P_{B0}}{P_{A0} \cdot P_{B1}}$$

The **odds ratio** (ψ) is the statistic most often used to interpret results of logistic model fitting.

$$\text{ex: } y = \begin{cases} 1 & \text{lung cancer} \\ 0 & \text{no lung cancer} \end{cases}$$

$$x = \begin{cases} A & \text{lifetime smoker} \\ B & \text{never a smoker} \end{cases}$$

$$\hat{\psi} = \frac{.50 / .50}{.05 / .95} = 19$$

Interpretation: A lifetime smoker has odds of contracting lung cancer that are estimated to be 19 times that of a never smoker.

Statements about Odds Ratios

$$\hat{\psi} = \frac{\frac{p(y=1|A)}{p(y=0|A)}}{\frac{p(y=1|B)}{p(y=0|B)}} = \frac{.50 / .50}{.05 / .95} = 19$$

- **Always correct:** A lifetime smoker's odds of contracting lung cancer are 19 times greater than the odds for a never smoker.
- If (probability of lung cancer < .05), then, "The probability that a lifetime smoker will contract lung cancer is 19 times greater than that of a never smoker."
 - This is a **Relative Risk (RR)** interpretation!
 - $RR = p(y=1 | A) / p(y=1 | B)$
 - Example: $RR = 0.50 / 0.05 = 10$ (very different from 19!)

Logit Function and Relationship to Odds Ratios

$$\text{logit}(Y=1|X)=\log\left(\frac{p(Y=1|X)}{p(Y=0|X)}\right)$$

$$\text{e.g. } \text{logit}(Y=1|X=A)=\log\left(\frac{p(Y=1|A)}{p(Y=0|A)}\right)=\log\left(\frac{.50}{.50}\right)=0.000$$

$$\text{logit}(Y=1|X=B)=\log\left(\frac{p(Y=1|B)}{p(Y=0|B)}\right)=\log\left(\frac{.05}{.95}\right)=-2.944$$

$$\psi(A:B)=\exp\left\{\log\left(\frac{p(Y=1|A)}{p(Y=0|A)}\right)\right\}\div\exp\left\{\log\left(\frac{p(Y=1|B)}{p(Y=0|B)}\right)\right\}$$

$$\psi(A:B)=\exp\left\{\log\left(\frac{p(Y=1|A)}{p(Y=0|A)}\right)-\log\left(\frac{p(Y=1|B)}{p(Y=0|B)}\right)\right\}$$

$$\psi(A:B)=\exp\{\text{logit}(Y=1|X=A)-\text{logit}(Y=1|X=B)\}$$

$$\psi(A:B)=\exp\{0.000-(-2.944)\}=19$$

Jumping Ahead

What if we could identify a linear model to estimate the logits:

$$g(E(Y)) = \underline{g(p(Y = 1|X))} = \beta_0 + \beta_1 X$$

if we choose $g(E(Y)) = \text{logit}(p(Y = 1 | X))$

$$\text{logit}(p(Y = 1 | X)) = \beta_0 + \beta_1 X$$

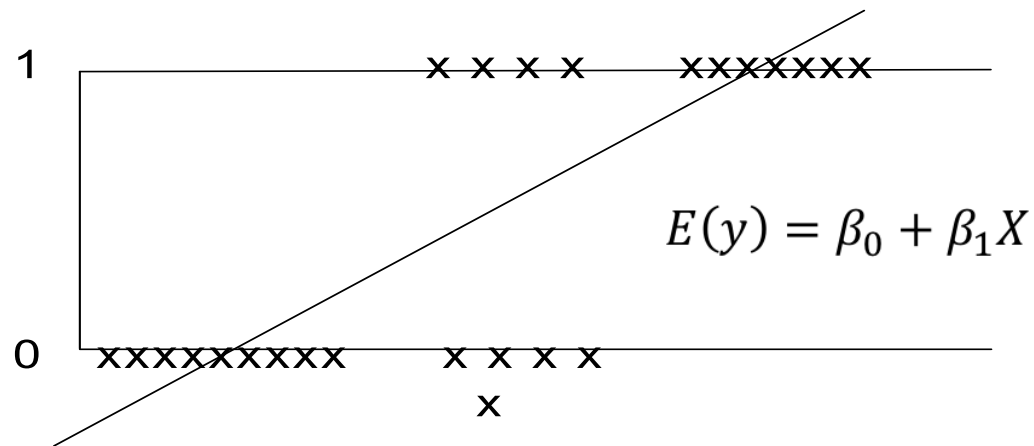
$$\text{e.g. define: } X = \begin{cases} 1 & \text{if smoker (A)} \\ 0 & \text{if not (B)} \end{cases}$$

then, from above:

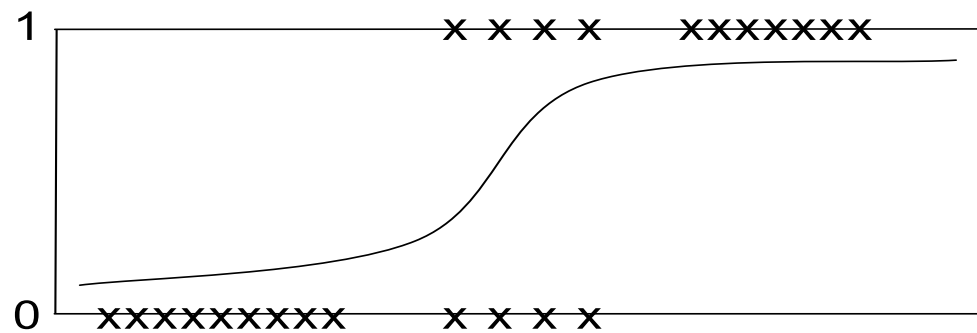
$$\begin{aligned} \psi &= \frac{\exp[\text{logit}(P_{1|A})]}{\exp[\text{logit}(P_{1|B})]} \\ &= \frac{\exp(\beta_0 + \beta_1)}{\exp(\beta_0)} \\ &= \exp(\beta_1) \end{aligned}$$

How Do We Model These Data?

linear in $E(Y)$? (OLS)



non-linear in $E(Y)$?

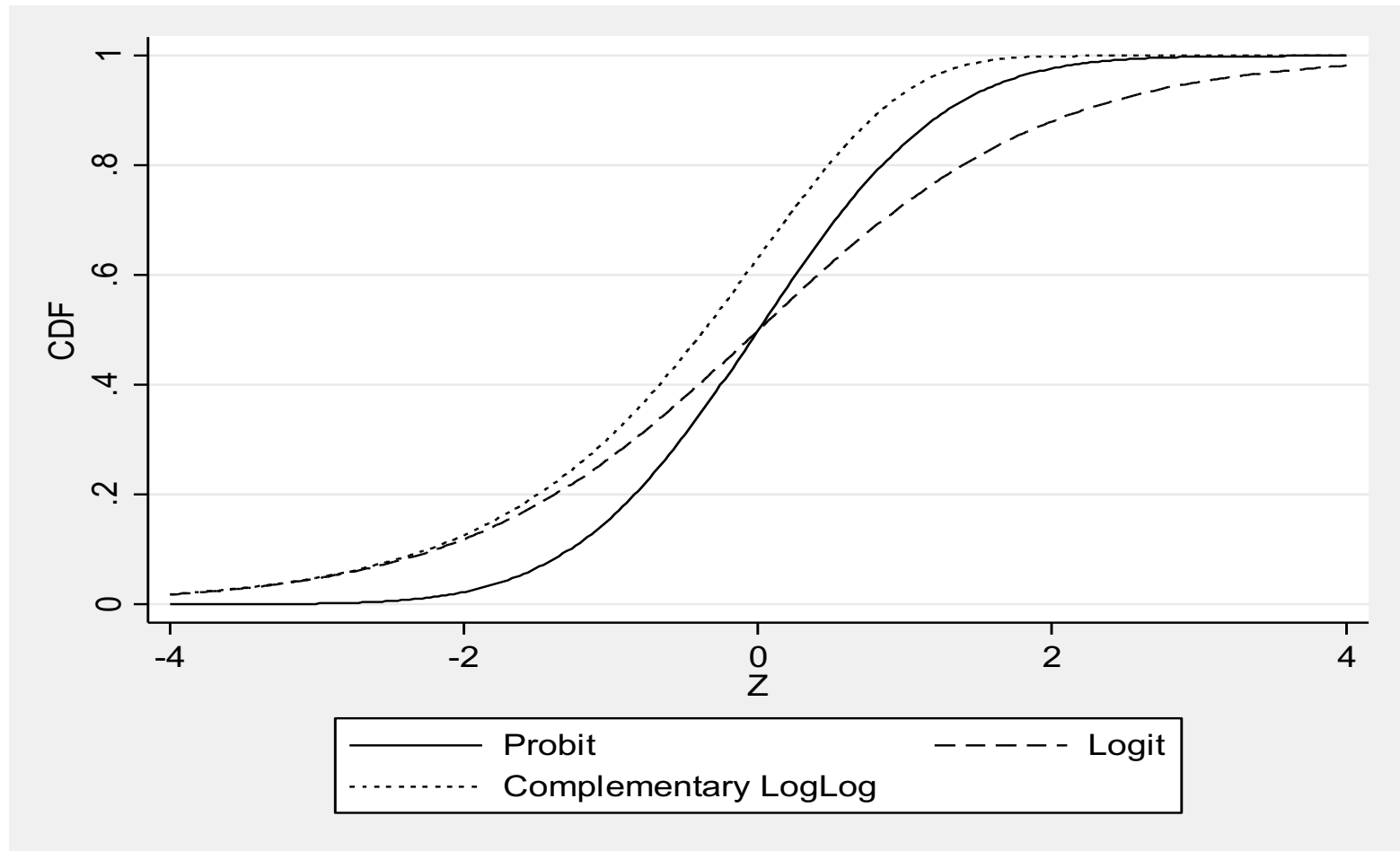


Generalized Linear Model: Logistic Regression

$$\begin{aligned} g(E(Y \mid \mathbf{x})) &= g(\pi(\mathbf{x})) = \text{logit}(\pi(\mathbf{x})) \\ &= \ln\left(\frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})}\right) = X\beta \end{aligned}$$

$$\hat{\pi}(\mathbf{x}) = g^{-1}(g(\hat{\pi}(\mathbf{x}))) = \frac{\exp(X\hat{\beta})}{1 + \exp(X\hat{\beta})}$$

Link Functions, $g(Z)$, for Binary Data



GLMs for Binary Data

- Three primary choices of links:
 - Logit
 - Probit
 - Complementary Log-Log (C-L-L)
- Focus here on **Logit**; few differences in model inferences, coefficients for probit are scaled differently from logit and C-L-L.
- Probit is often used in economics and related disciplines.
- Complementary-log-log has application when predicted probabilities are near 0 or 1.
- Stata (PROC SURVEYLOGISTIC also features these link functions):
 - svy: logit and svy: logistic
 - svy: probit
 - svy: cloglog

Logistic Regression Model (SRS Setting)

$$\text{Likelihood} = \prod_{i=1}^n \pi(x_i)^{Y_i} \cdot [1 - \pi(x_i)]^{1-Y_i}$$

$$\text{where: } \pi(x_i) = \frac{\exp(x_i\beta)}{1 + \exp(x_i\beta)}$$

Parameter estimates: $\hat{\beta}_{MLE} = \{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p\}$.

An iterative algorithm is used to obtain MLEs
(Newton-Raphson method)

Variance of Parameter Estimates: $\hat{V}(\hat{\beta}) = I^{-1}(\beta)|_{\beta = \hat{\beta}}$
the inverse of the “information matrix” $I(\beta)$, evaluated at the
maximum likelihood estimate

Confidence Intervals for Parameter Estimates (SRS)

$$CI(\beta) = \hat{\beta} \pm 1.96 \times se(\hat{\beta}) \text{ for } \alpha = .05$$

$$\hat{\psi}_j = e^{\hat{\beta}_j}$$

$$CI(\psi_j = e^{\beta_j}):$$

$$\text{Lower - } e^{\hat{\beta}_j - 1.96 \times se(\hat{\beta}_j)}$$

$$\text{Upper - } e^{\hat{\beta}_j + 1.96 \times se(\hat{\beta}_j)}$$

$\hat{\psi}_j$ = multiplicative change in odds for a one-unit change in X_j , with all other $X_{j \neq k}$ held constant (e.g., 1.5 implies that the odds are expected to increase by 50%).

Testing Significance of Effects in Fitted Models (SRS)

Likelihood ratio test:

Under $H_0: \beta_j = 0$

$$G = -2\ln\left(\frac{\text{likelihood} - \text{reduced model}}{\text{likelihood} - \text{full model}}\right) \sim X_q^2,$$

Chi-square with q degrees of freedom

$q = \#$ parameters in full model - $\#$ parameters in reduced model

Odds Ratios in Multivariate Models

- Single X, no interactions:

$$\hat{\psi}_{x=X_k} = e^{\hat{\beta}_k}$$

= the odds ratio for $y=1$,
comparing X_k+1 to X_k ,
holding all other variables
fixed at $\tilde{x}_{j \neq k} = \tilde{X}_{j \neq k}$

Odds Ratios in Multivariate Models

Multiple X variables (predictors) and/or Interactions:

Given $\tilde{X}_1, \tilde{X}_2$ – two covariate "patterns"

1) Evaluate: $\text{logit}(y=1 | \tilde{X}_1), \text{logit}(y=1 | \tilde{X}_2)$

2) Compute difference:

$$\Delta = \text{logit}(y=1 | \tilde{X}_1) - \text{logit}(y=1 | \tilde{X}_2)$$

3) Estimate: $\hat{\psi}_{X1 \text{ vs. } X2} = e^{\Delta}$

Goodness of Fit, Model Diagnostics

- Criteria (Hosmer et al., 2013):
 - 1) $(y - \hat{\pi})$ should be small and not irregular over x
 - 2) Individual $(y_j - \hat{\pi}_j)$ should not have extreme influence
- Goodness of Fit
 - Hosmer-Lemeshow Test
 - Pearson, Deviance
 - ROC, Area under the curve
 - Squared correlation between (Y, π)
 - Pseudo R^2 (not equivalent to R^2 in linear regression, and only approximate; use to compare models only)
- Influence
 - “Hat” values for logistic regression
 - “Cook’s D-like” measures
 - Change in X^2 due to deletion of cases or “patterns” of cases with identical x ’s.

Logistic Regression Analysis for Complex Sample Survey Data

Recall SRS Results:

$$\ln\left(\frac{\pi(X_i)}{1-\pi(X_i)}\right) = \beta_0 + \beta_1 X_i$$

multiple logistic regression:

$$\pi(x_i) = \frac{e^{x_i, \beta}}{1 + e^{x_i, \beta}}$$

$$\underline{\beta} = \{\beta_0, \beta_1, \dots, \beta_p\}$$

$$\underline{x} = \{x_1, x_2, \dots, x_p\}$$

$$l(\beta) = \prod_{i=1}^n \pi(x_i)^{Y_i} (1 - \pi(x_i))^{1 - Y_i}; Y_i \text{ are distributed as } B(1, \pi(x_i))$$

- ✓ $\hat{\beta}$ by maximum likelihood
- ✓ se $(\hat{\beta})$ by $I^{-1}(\beta) | \beta = \hat{\beta}$
- ✓ LRT to test model effects

What's Different from the SRS Setting?

Complex Sample Survey Data:

$$\ln\left(\frac{\pi(X_i)}{1-\pi(X_i)}\right) = B_0 + B_1 X_i$$

$$\pi(X_i) = \frac{e^{X_i B}}{1 + e^{X_i B}} \quad B = \{B_0, B_1, \dots, B_p\}$$

$L(B)$ is different from $L(\beta)$ in the SRS setting, due to weights, stratification, and cluster sampling!

$L(B) = ?$

- $\hat{B} \neq \hat{\beta}$
- $\hat{B}$ estimated?
- $se(\hat{B}) \neq I^{-1}(B) | B = \hat{B}$
- $se(\hat{B})$ estimated?
- Hypothesis tests? Goodness of fit?

Iterative Pseudo-Likelihood Method for Logistic Regression (Binder, 1981, 1983)

Recall: $L(B)_{(\text{complex})}$ is different from $L(\beta)_{(\text{SRS})}$

Approach:

- 1) Estimate B as we would β with iterative maximum likelihood estimation methods (incorporating the weights).
- 2) Derive the $\hat{Var}(\hat{B})$ as a matrix function of the “score equations,” which are linear functions [we know how to estimate complex sample variances for linear functions of totals]. Simpler alternatives: JRR and BRR.
- 3) Use Wald-type statistics for making inferences about B .

Pseudo-Maximum Likelihood for Logistic Regression

Pseudo ln(Likelihood):

$$\ln PL(B) = \sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i} y_{h\alpha i} \cdot \ln(\pi(x_{h\alpha i})) + \sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i} (1 - y_{h\alpha i}) \cdot \ln(1 - \pi(x_{h\alpha i}))$$

where:

$$\pi(x_i) = e^{x_i^B} / (1 + e^{x_i^B}), \quad \hat{\pi}(x_i) = e^{x_i^b} / (1 + e^{x_i^b})$$

and b = the vector of coefficient estimates that solves:

$$S(B) = \frac{\delta \ln PL(B)}{\delta B} \Big|_{B=b} = 0 =$$

$$= \sum_h \sum_{\alpha} \sum_i w_{h\alpha i} (y_{h\alpha i} - \hat{\pi}(x_{h\alpha i})) x'_{h\alpha i} = \sum_h \sum_{\alpha} \sum_i s_{h\alpha i}$$

Variance-Covariance Matrix for Estimated Parameters: Linearization

$$\begin{aligned} \mathbf{J} &= - \left[\frac{\delta^2 \ln PL(\mathbf{B})}{\delta^2 \mathbf{B}} \right] \bigg|_{\mathbf{B} = \hat{\mathbf{B}}} \\ &= \sum_h \sum_{\alpha} \sum_i \mathbf{x}'_{h\alpha i} \mathbf{x}_{h\alpha i} w_{h\alpha i} \hat{\pi}_{h\alpha i}(\mathbf{B})(1 - \hat{\pi}_{h\alpha i}(\mathbf{B})) \end{aligned}$$

$$\text{Var}(\hat{\mathbf{B}}) = (\mathbf{J}^{-1}) \text{var}[S(\hat{\mathbf{B}})](\mathbf{J}^{-1})$$

Variance-Covariance Matrix for Estimated Parameters: Linearization

$$\text{var}[S(\hat{\mathbf{B}})] = \frac{n-1}{n-(p+1)} \sum_{h=1}^H \frac{a_h}{(a_h-1)} \sum_{\alpha=1}^{a_h} (s_{h\alpha} - \bar{s}_h)' (s_{h\alpha} - \bar{s}_h)$$

which for n large is:

$$\text{var}[S(\hat{\mathbf{B}})] \cong \sum_{h=1}^H \frac{a_h}{(a_h-1)} \sum_{\alpha=1}^{a_h} (s_{h\alpha} - \bar{s}_h)' (s_{h\alpha} - \bar{s}_h)$$

where for the logistic link :

$$s_{h\alpha} = \sum_{i=1}^{n_{h\alpha}} s_{h\alpha i} = \sum_{i=1}^{n_{h\alpha}} w_{h\alpha i} (y_{h\alpha i} - \hat{\pi}_{h\alpha i}(\mathbf{B})) \mathbf{x}'_{h\alpha i} ; \text{ and}$$

$$\bar{s}_h = \frac{1}{a_h} \sum_{\alpha=1}^{a_h} s_{h\alpha}$$

Hypothesis Testing

$$B' = [B_1, \dots, B_p]$$

$$H_o: CB = 0$$

$$X_{Wald}^2 = [C\hat{B}]' [CV\hat{ar}(\hat{B})C']^{-1} [C\hat{B}]$$

$$= \frac{\text{contrast "squared"}}{\text{variance of contrast}}$$

Under H_o : X_{Wald}^2 is distributed as χ_q^2

where q = the rank of C

Hypothesis Testing: 1

$$B' = B_1, B_2, B_3$$

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

CB tests $H_0: B_2 = 0$
d.f. = rank(C) = 1

Hypothesis Testing: 2

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

CB tests $H_0: B_2 = B_3 = 0$
d.f. = rank(C) = 2

Hypothesis Testing: 3

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

CB tests $H_0: B_2 - B_3 = 0$
d.f. = rank(C) = 1

Hypothesis Testing in Stata

Tests of individual coefficients :

$$H_0 : 1 \cdot B_j = 0$$

$$t = \frac{b_j - 0}{se(b_j)}$$

under $H_0 \sim \text{Student } t$, with $df = \#clusters - H$

Tests of multiple coefficients :

$$H_0 : \underline{B}_{(2)} = \{B_{q+1}, B_2, \dots, B_p\} = \dots, 0\}$$

$$F_{Wald} = \frac{\chi_{Wald}^2}{(p - q)}$$

under $H_0 \sim F$ $df = (p - q, \#clusters - H)$

Wald Tests in Stata

- First, code the model fit in the `svy: regress` (or `svy: logit`, `svy: poisson`, etc.) regression program
- Follow the execution of the `svy: [ ]` regression command with the execution of a test statement
- e.g. (for ASDA Chapter 8 example), to test the null hypothesis that all of the parameters associated with AG4CAT are equal to zero, use this test statement:

```
test 2.ag4cat 3.ag4cat 4.ag4cat
```

Software Procedures for Logistic Regression Analysis of Survey Data

- **Stata:** the `svy: logit` and `svy: logistic` commands
 - `svy: logistic` defaults to odds ratio output; **coef** option yields logistic model parameter estimates
 - `svy: logit` defaults to log-odds (B) output; **or** option yields odds ratios
 - “i.” prefix defines categorical predictors
 - Default to lowest alphanumeric category for reference
 - Change reference category for categorical predictors using `ib#`.
 - Post-estimation **test** statement for Wald tests of multi-parameter hypotheses

Example: Logistic regression Analysis of NCS-R data: Specifying the Preliminary Model

- In this example, we assess the significance of potential predictors of having lifetime major depression for adults greater than 17 years of age.
- **Dependent Variable**: (MDE: 1=Yes, 0=No)
- **Predictor Variables**:
 - AGE4CAT (1=18-29; 2=30-44; 3=45-49; 4=60+)
 - SEX (1 = Male; 2 = Female)
 - ALD (1 = alcohol dependent; 0= not ALD)
 - EDCAT (1=<12; 2=12; 3=13-15; 4=16+)
 - MARCAT (1=Married; 2=Previously; 3=Never)

Example: Logistic regression Analysis of NCS-R data: Examining Bivariate Associations.

Stata Code:

```
svyset seclustr [pweight = ncsrwtlg], strata(sestrat)
svy: tab ag4cat mde, row
svy: tab sex mde, row
svy: tab ald mde, row
svy: tab ed4cat mde, row
svy: tab mar3cat mde, row
```

Example: Logistic regression Analysis of NCS-R data: Examining Bivariate Associations.

Predictor	Rao-Scott <i>F</i> -test	Category	% with MDE (SE)
AGE4CAT	$F(2.76,115.97) = 26.39$ $p < 0.01$	18-29	18.4% (0.9)
		30-44	22.9% (1.1)
		45-59	22.3% (1.3)
		60+	11.1% (1.0)
SEX	$F(1,42) = 44.83$ $p < 0.01$	Male	15.3% (0.9)
		Female	22.6% (0.7)
ALD	$F(1,42) = 120.03$ $p < 0.01$	Yes	45.2% (0.3)
		No	17.7% (0.7)
ED4CAT	$F(2.90,121.93) = 4.30$ $p < 0.01$	<12 yrs	16.3% (1.2)
		12 yrs	18.6% (0.8)
		13-15 yrs	21.3% (1.0)
		16+ yrs	19.7% (1.1)
MAR3CAT	$F(1.90,79.74) = 11.08$ $p < 0.01$	Married	17.3% (0.7)
		Previously	23.9% (1.5)
		Never	19.4% (1.1)

Example: Logistic regression Analysis of NCS-R data: Fit preliminary model.

Stata Code:

```
svy: logistic mde i.ag4cat ib2.sex ald i.ed4cat i.mar3cat  
svy: logistic mde i.ag4cat ib2.sex ald i.ed4cat i.mar3cat, coef
```

```
svy: logit mde i.ag4cat ib2.sex ald i.ed4cat i.mar3cat
```

Estimated odds ratios and 95% CIs can be generated in svy: logit by adding the or option:

```
svy: logit mde i.ag4cat ib2.sex ald i.ed4cat i.mar3cat, or
```

Other software systems (e.g., SAS PROC SURVEYLOGISTIC) will output both the estimated logistic regression coefficients (and standard errors) and the corresponding odds ratio estimates.

Example: Logistic Regression Analysis of NCS-R data: Fit Preliminary Model.

Predictor*	Category	$\hat{B}$	$se(\hat{B})$	t	$P(t_{42} > t)$
INTERCEPT		-1.583	0.121	-13.12	<0.001
AGE4CAT	30-44	0.255	0.122	2.71	0.100
	45-59	0.206	0.113	2.26	0.029
	60+	-0.676	0.072	-4.78	< 0.001
SEX	Male	-0.577	0.138	-7.48	< 0.001
ALD	Yes	1.424	0.640	9.24	< 0.001
ED4CAT	12	0.079	0.105	0.82	0.418
	13-15	0.231	0.117	2.48	0.017
	16+	0.163	0.130	1.47	0.148
MAR3CAT	Previously	0.486	0.139	5.69	<0.001
	Never	0.116	0.121	1.07	0.290

* Source: NCS-R, $n = 5,692$, adjusted Wald test for all parameters: $F(10,33) = 28.07, p < 0.001$.
Reference categories for categorical predictors are: AGE4CAT (18-29); SEX (Female); ALD (No); ED4CAT (<12 yrs); MAR3CAT (Married).

Example: Logistic Regression Analysis of NCS-R data: Interpret Model Results as Odds Ratio Estimates.

Predictor*	Category	$\hat{\theta}$	95% CI for $\hat{\theta}$
AGE4CAT	30-44	1.29	(1.067, 1.562)
	45-59	1.23	(1.022, 1.479)
	60+	0.51	(0.383, 0.677)
SEX	Male	0.56	(0.480, 0.656)
ALD	Yes	4.15	(3.042, 5.668)
ED4CAT	12	1.08	(0.890, 1.316)
	13-15	1.26	(1.044, 1.519)
	16+	1.18	(0.941, 1.471)
MAR3CAT	Previously	1.63	(1.369, 1.932)
	Never	1.12	(0.903, 1.396)

**Example: Logistic Regression Analysis of NCS-R data:
Wald Tests of Multi-parameter Predictors.**

test 2.ag4cat 3.ag4cat 4.ag4cat

test 2.mar3cat 3.mar3cat

test 2.ed4cat 3.ed4cat 4.ed4cat

Categorical Predictor	F-test Statistic	$P(\mathcal{F} > F)$
AGE4CAT	$F_{(3,40)} = 19.03$	< 0.001
ED4CAT	$F_{(3,40)} = 2.13$	0.112
MAR3CAT	$F_{(2,41)} = 16.60$	< 0.001

**Example: Logistic Regression Analysis of NCS-R data:
Test overall goodness of fit of preliminary model.**

- Use Archer and Lemeshow's (2006, 2007) design-adjusted test to assess the goodness of fit of this initial model.
- estat gof (post-estimation command)
- The resulting design-adjusted F -statistic reported in the Stata Results window is equal to $F_{A-L} = 1.229$, with a p -value of 0.310. This suggests that the null hypothesis that the model fits the data well is not rejected. We therefore have confidence moving forward that the fit of this initial model is reasonable.

**Example 2: Logistic regression analysis of NCS-R ALD.
Compare Logit, Probit, CLL Link Models.**

```
svy: logit ald i.ag4cat ib2.sex i.ed4cat i.mar3cat
```

```
svy: probit ald i.ag4cat ib2.sex i.ed4cat i.mar3cat
```

```
svy: cloglog ald i.ag4cat ib2.sex i.ed4cat i.mar3cat
```

Example 2: Logistic regression analysis of NCS-R ALD. Compare Logit, Probit, CLL Link Models.

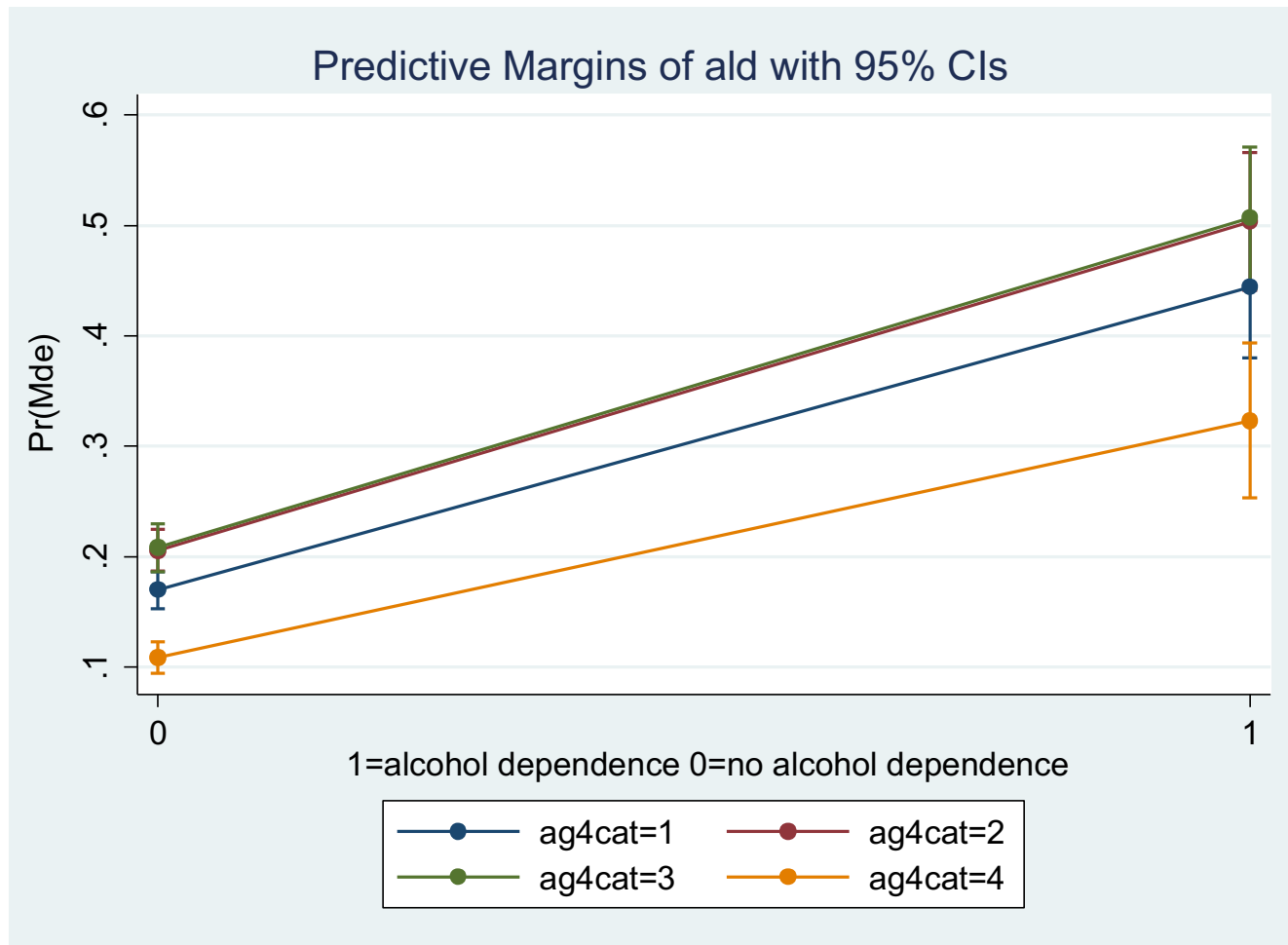
Predictor*	Category	Logistic			Probit			Complementary L-L		
		$\hat{B}$	$se(\hat{B})$	p	$\hat{B}$	$se(\hat{B})$	p	$\hat{B}$	$se(\hat{B})$	p
Intercept		-3.124	0.225	<0.01	-1.719	0.105	<0.01	-3.140	0.218	<0.01
SEX	Male	0.997	0.119	<0.01	0.471	0.056	<0.01	0.965	0.115	<0.01
AGE4CAT	30-44	0.146	0.178	0.416	0.065	0.084	0.444	0.143	0.171	0.408
	45-59	-0.051	0.144	0.726	-0.034	0.067	0.609	-0.045	0.140	0.748
	60+	-1.120	0.212	<0.01	-0.531	0.093	<0.01	-1.083	0.209	<0.01
ED4CAT	12 yrs	-0.268	0.194	0.173	-0.124	0.095	0.200	-0.260	0.185	0.167
	13-15 yrs	-0.264	0.176	0.141	-0.124	0.085	0.152	-0.256	0.169	0.137
	16+ yrs	-0.736	0.197	<0.01	-0.339	0.092	<0.01	-0.713	0.190	<0.01
MAR3CAT	Previously	0.517	0.142	<0.01	0.255	0.069	<0.01	0.494	0.136	<0.01
	Never	0.065	0.169	.0701	0.039	0.077	0.616	0.060	0.164	0.713

Plotting Predicted Marginal Probabilities and Effects in Stata

- Stata offers extremely easy-to-use post-estimation commands for calculation and plotting of marginal predicted probabilities based on fitted models
- Default calculation: compute a model-based predicted probability for everyone in the data set as if they all belonged to the same subgroup, and average the predictions
- One can plot marginal predicted probabilities for different subgroups, or average marginal effects (i.e., expected changes in predicted probabilities associated with a one-unit increase in a given predictor)
- The next few slides present some of the examples illustrated in Chapter 8 of ASDA (Second Edition)

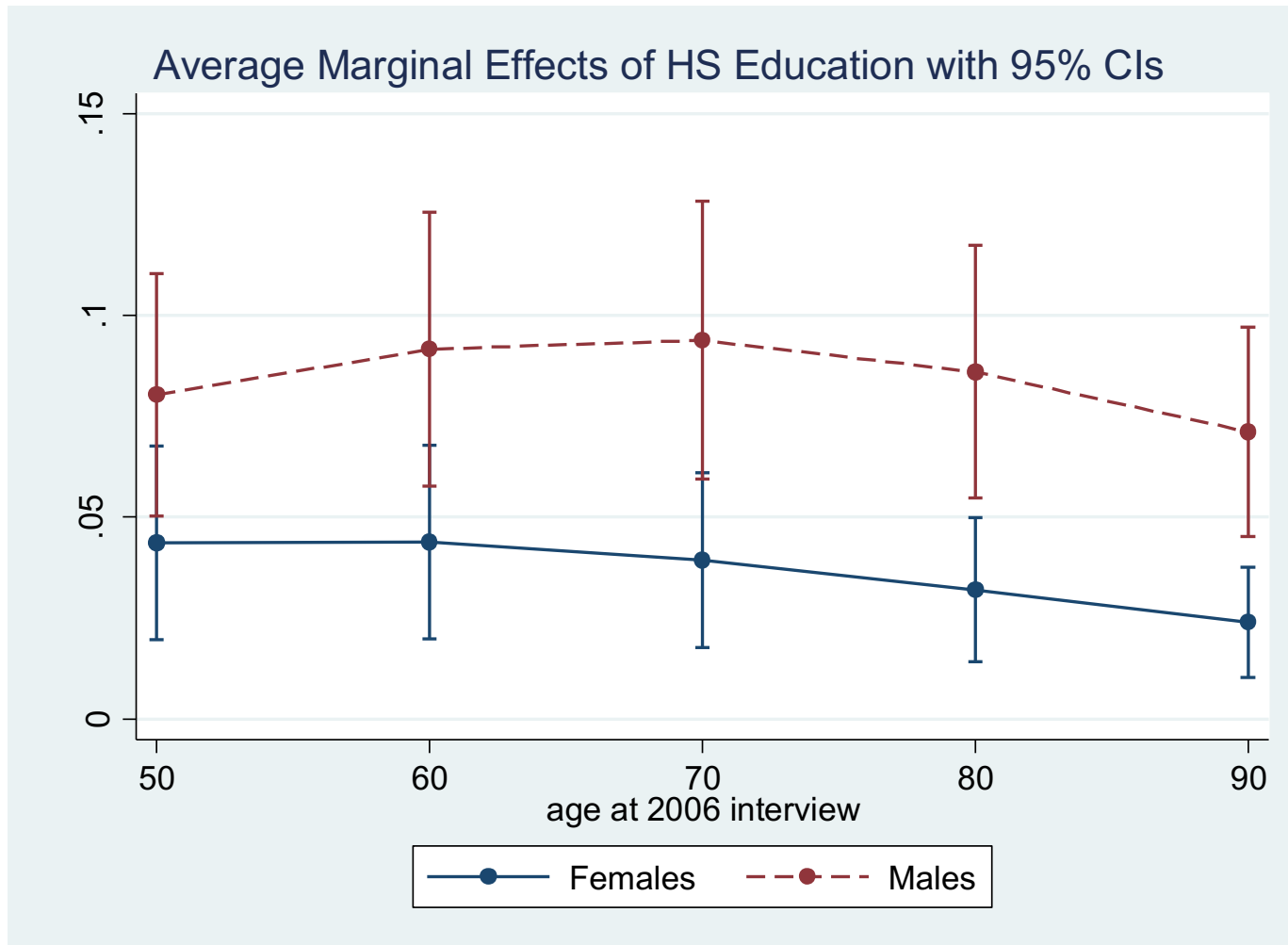
Plotting Marginal Predicted Probabilities in Stata

```
svy: logit mde i.ald i.ag4cat  
margins ald, by(ag4cat)  
marginsplot
```



Plotting Average Marginal Effects in Stata

```
margins, dydx(2.edcat3) by(male) at(kage=(50(10)90))  
marginsplot
```



Plotting Average Marginal Effects in Stata

```
margins, dydx(ald) by(ag4cat)  
marginsplot
```

